

Speeding up 6-DoF Grasp Sampling with Quality-Diversity

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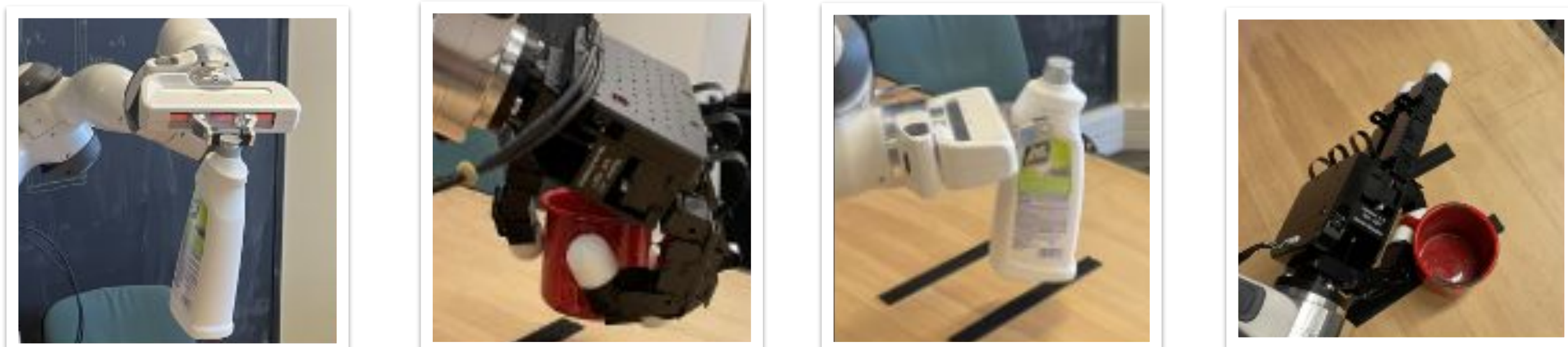
Abstract

This paper proposes an efficient method to generate simulated datasets for grasping based on quality-diversity algorithms.

Introduction

How does a robot grasp an object?

- Grasp = select a specific **configuration of the robot end effector**

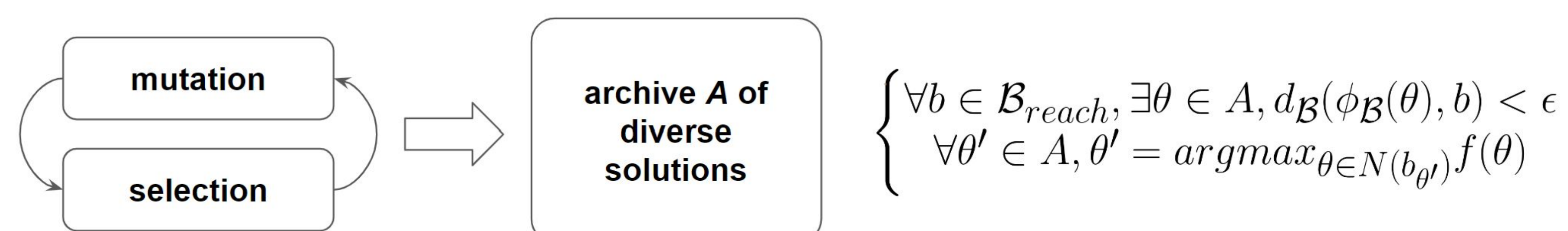


How to learn to grasp in robotics?

- Objective : teach a robot how to grasp any object
- Problem: demonstrations are challenging to obtain
- Solution proposed : generate a **large dataset for grasping with quality-diversity methods**

What are Quality-Diversity methods

- **Optimization algorithms**
- Generate a set of **diverse** and high-performing solutions
- **Evolutionary algorithm** = selection-mutation loop



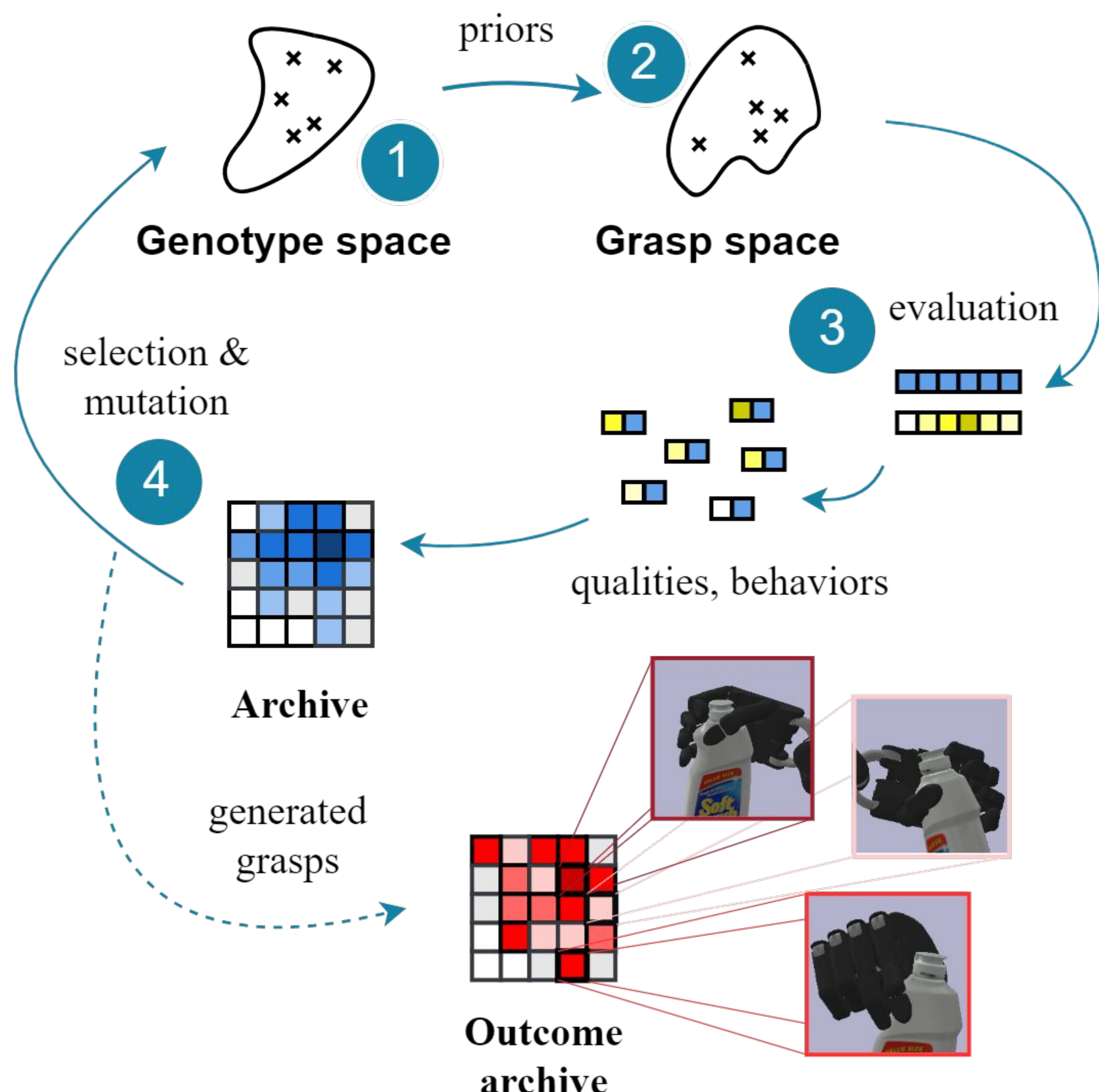
Problematic

How to speed-up 6-DoF grasp sampling using Quality-Diversity methods ?

Method

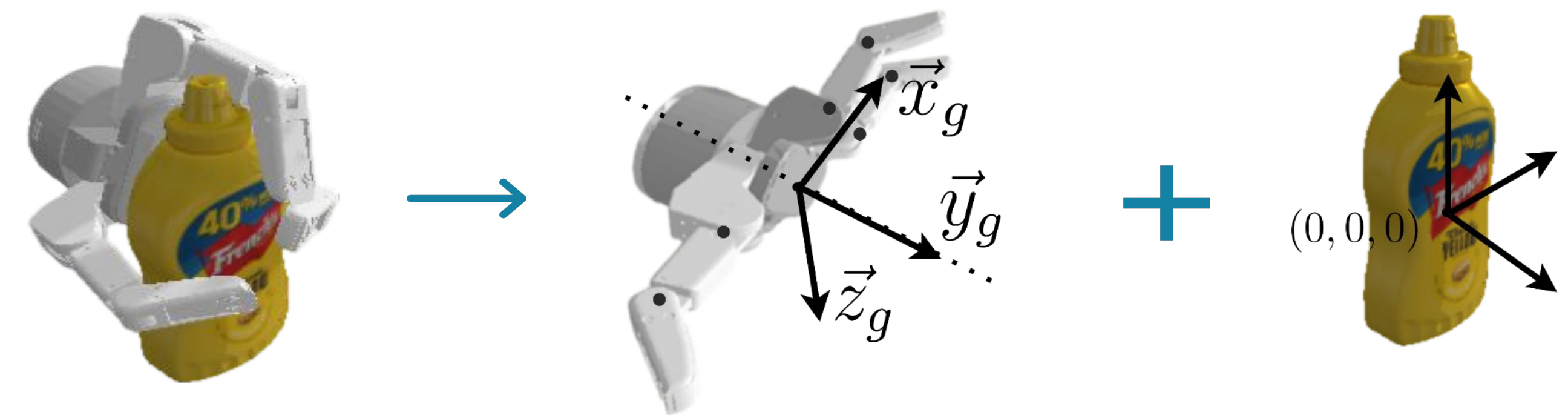
Concept : Quality-Diversity algorithms to generate grasps

- Generation : **iterative generative loop**



1 Sample 6-DoF grasp

- Sample = spatial configuration + internal link configuration
- **6-DoF grasp sample** = $\theta = (x_g, y_g, z_g, d, \nu, \xi, \omega)$



2 Physical priors

- Antipodal prior
- Approach prior
- Contact prior

3 Evaluation

- **Quality of a grasp** = robustness
- **Diversity of a grasp** = novelty

4 Archive storage and selection-mutation

- Successful grasps are stored
- Top-performing grasps are selected



Experiments

Comparison of methods

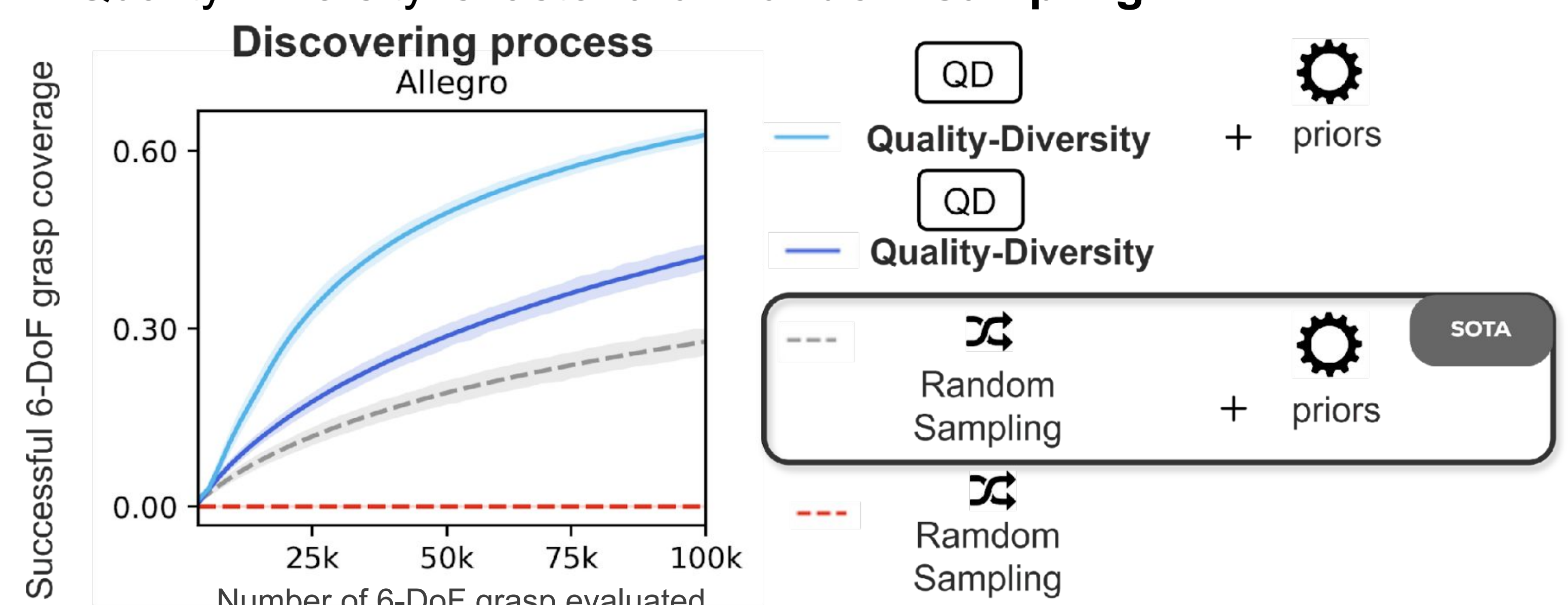
- 4 end-effectors, 10 objects
- Random sampling VS Quality-Diversity sampling



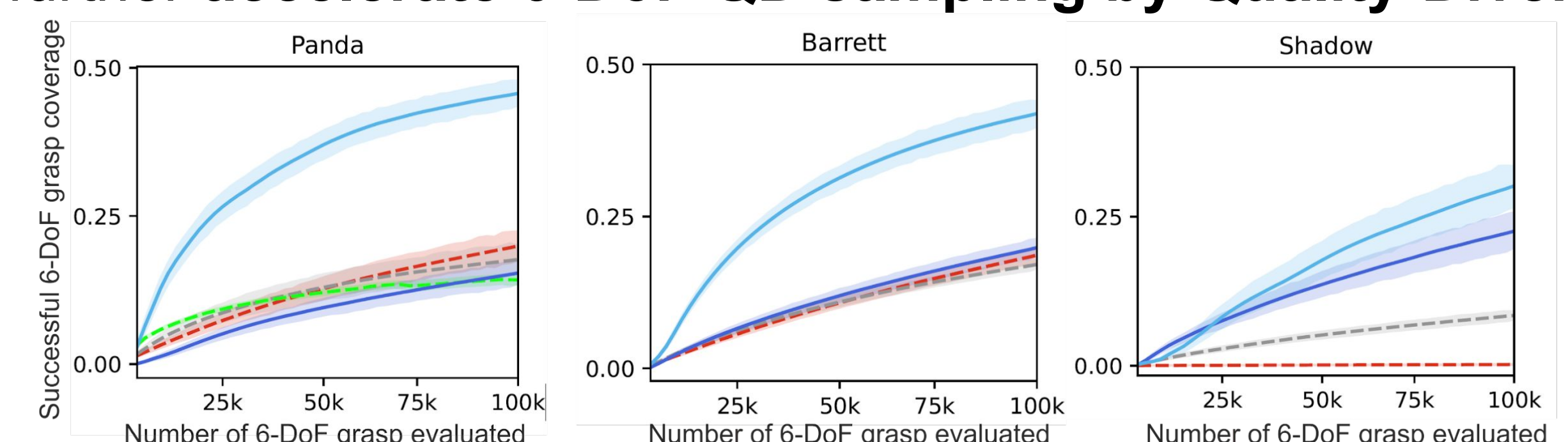
Results

Quality-Diversity Speeds-up 6-DoF grasp sampling

- Quality-Diversity is faster **than random sampling**

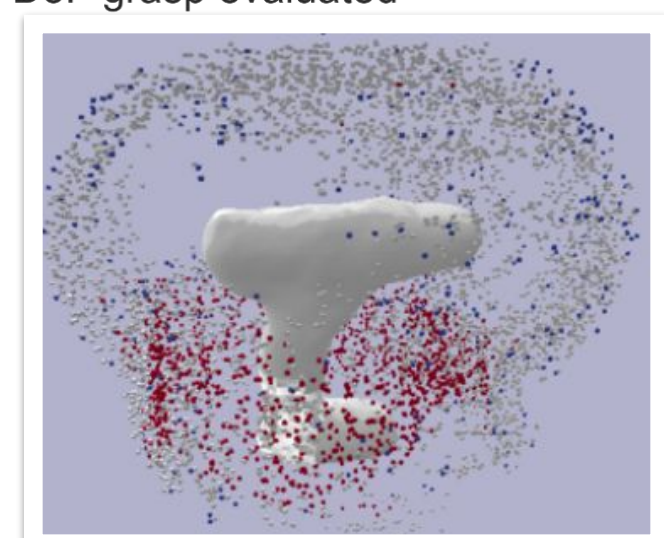


- Priors further **accelerate 6-DoF QD sampling by Quality-Diversity**



Diversity in the grasp coverage

- Grasps covering the entire object surface
- Transferability : 95% Panda gripper, 72% allegro hand



Discussions

- Benefits : fast and automatic generation, diverse solutions
- Limitation : GPU parallelization, force intensity analysis

Contact and website

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Project webpage : <https://qdgrasp.github.io/>